THE LAST SHIFT

Project 4 Swap Day | Team 3 | October 23, 2026

Name: Rocklyn (Rocky) Clarke
Mechanic owned: Shifting Environment
Received mechanics: Team 2
Public repo: https://github.com/notziking8/Project-4.git
My commit hash: 2272fc95b1b238dd42dcacaa95d10c8641b62142

Part 1 - Read it without help (30 points)

The answers below separate the handoff rules from our implementation guesses. We keep all four received mechanics. Conflicts are recorded rather than silently treated as clear. We will not contact Team 2 or Team 4 about the project before the final playtest ends.

1. Utility / Inventory - HAD TO GUESS

Question: Does armor use 2/3/4 slots, or does every suit use 2?

Build answer: Use the Utility section's type-specific costs: light = 2, medium = 3, heavy = 4; weapons = 1 slot as an implementation assumption; relics = 1. Capacity is exactly 10 with no upgrades. Equipped items continue to occupy inventory slots. This resolves the contradictory Armor wording as our team build interpretation.

Question: Can equipment be changed outside combat, or only at checkpoints?

Build answer: Use the stricter Utility rule: equip, switch, and voluntarily drop armor or relics only while inside an activated checkpoint. For consistency, weapon swaps and voluntary weapon drops follow the same restriction. Pickups with E are allowed anywhere if space is available; pickups never auto-equip.

Question: What happens when a pickup will not fit?

Build answer: Reject the entire pickup without removing it from the world. There is no partial pickup or automatic dropping. An item remains at its location until collected or the level is restarted. An equipped suit and all backup suits count toward the same capacity.

Question: What are relic effects and how are they used during combat?

Build answer: Healing Relics are single-use emergency medical capsules. Each capsule occupies one inventory slot, including duplicates; they do not stack into one slot. Press R to consume one carried capsule and immediately restore 25 HP, capped at 100 HP. Use is allowed during combat and has a 5-second cooldown. At full health, with no capsule, during cooldown, or while dead, R does nothing and consumes nothing. A successful use removes exactly one capsule and frees its slot; consumption is not dropping equipment and does not require a checkpoint. The player starts with two capsules. Place two additional capsules in each sector, collected with E using the same capacity rules; collected capsules do not respawn after death. Unused capsules persist through death, consumed ones stay consumed, and the cooldown resets on respawn. A full level restart resets all capsules. These rules complete the undefined relic behavior within Utility / Inventory.

Part 1 - Armor interpretation

2. Armor Types - HAD TO GUESS

Clear rules preserved: One equipped suit at a time; no durability or maximum-health increase. Light gives 15% damage reduction and +10% speed. Medium gives 35% damage reduction and 25% elemental resistance. Heavy gives 60% damage reduction, knockback immunity, and -15% speed.

Question: How do general damage reduction and elemental resistance combine?

Build answer: Apply general reduction first, then medium armor's elemental reduction only to elemental damage. A 20-point physical hit becomes 17 / 13 / 8 with light / medium / heavy armor. A 20-point elemental hit becomes 9.75 with medium armor: 20 x 0.65 x 0.75. Keep fractional health internally.

Question: What is the base movement speed, and where does elemental resistance matter?

Build answer: Use base speed 4 units/second: light = 4.4, medium = 4, heavy = 3.4. As a documented content assumption, some Light enemies deal electrical damage using their unchanged 10 damage, 2-unit attack range, and 1.5-second interval. Other attacks are physical. This supplies an elemental damage type without a separate hazard system or new player action.

Question: What does knockback immunity prevent?

Build answer: Ordinary enemy hits push the player 0.5 unit away from the attacker, stopped by collision. Heavy armor prevents that displacement. Knockback adds no extra damage and does not stun. The distance is an implementation assumption because the handoff supplies no default knockback value.

Question: When does combat end, and does that allow an armor swap?

Build answer: Combat ends when no enemy is pursuing or attacking the player. It does not unlock swapping under our checkpoint-only interpretation. A checkpoint cannot be activated while combat is active; entering it during combat does not activate it or enable swapping.

Question: How do death and backup armor interact with the inventory?

Build answer: At zero health, lose the attempt and respawn at the most recent checkpoint with 100 health and the same equipped armor. Remove every unequipped backup suit from inventory and place it at the death location. Retain weapons and relics because the handoff only explicitly drops backup armor. Recovered items require E and enough capacity.

Question: Are locked lockers or miniboss rewards required?

Build answer: Preserve all three listed sources. Spawn one armor reward at each newly activated checkpoint. Security lockers unlock when their sector is cleared using the existing enemy-clearance condition. Designate selected Heavy enemies as elite minibosses, retaining their existing stats and assigning armor as their guaranteed drop. Collect all armor with E and the same capacity checks. These are acquisition-content assumptions; no keys, lockpicking action, or third enemy type is added.

Checkable inventory example

A weapon (1), light armor (2), medium backup armor (3), and four healing relics (4) fill all 10 slots. At 60 HP, pressing R restores health to 85, consumes one relic, and reduces occupied slots to 9. A further relic pickup can then fit. At 100 HP, R consumes nothing. Voluntary relic drops still require a checkpoint.

Part 1 - Environment and enemy interpretation

3. Shifting Environment - HAD TO GUESS

Clear rules preserved: All eligible objects attempt to shift every 30 seconds between two fixed positions no more than 5 feet apart. F freezes one aimed-at object within 10 feet for 10 seconds; only one freeze can exist. A 5-second cooldown begins when it expires. A thawed object waits for the next global shift. Objects cannot hurt, overlap, or fully trap the player.

Question: How do feet translate to the enemy system's units?

Build answer: Use 1 Godot unit = 1 meter. Maximum shift = 1.524 units; flashlight range = 3.048 units. Enemy detection, attack, and leash distances remain 6, 2, and 10 Godot units.

Question: What if a shift would overlap the player or seal every route?

Build answer: Validate each proposed change against collision and a remaining walkable route before applying it. Skip any unsafe change until the next global 30-second event. Author both layouts and reachable mixed frozen layouts so a legal escape always exists. Use the same player-clearance checks for respawn placement.

Question: Can the freeze be refreshed, and what happens at an exact timer boundary?

Build answer: No refreshing or retargeting until both the 10-second freeze and following 5-second cooldown finish. F needs line of sight and does nothing on an invalid target. At a shared timestamp, process timer expirations first, then valid F inputs, then the environmental shift. A thawed object may shift at that event; a newly frozen object may not. The post-freeze cooldown prevents immediate re-freezing.

4. Enemy System - HAD TO GUESS

Clear rules preserved: Light: 50 HP, 10 damage, 4 units/second, 1.5 seconds between attacks. Heavy: 100 HP, 20 damage, 2 units/second, 3 seconds between attacks. Detect within 6 units, attack within 2, stop pursuit when the player is over 10 units from the enemy's starting position. Navigate around walls. Each defeat drops one weapon or armor item. Clear every enemy to advance.

Question: What weapon, damage, range, and starting health does the player have?

Build answer: Documented assumptions: 100 HP; one reusable weapon with 25 damage, a 0.5-second attack cooldown, 10-unit range, and no ammunition system. Left click fires at the first collision under the aim point; walls block shots. Start with this equipped weapon, light armor, and two Healing Relics, using 5 inventory slots total. Two hits defeat a Light enemy; four defeat a Heavy enemy.

Question: Can enemies attack or detect through walls, and what survives a death?

Build answer: Require line of sight for detection and attacks. First attack occurs after its stated attack interval, then repeats at that interval while valid. When leashed, return to the starting point without healing. Player death retains defeated enemies, remaining enemy HP, uncollected drops, and activated checkpoints. Restarting the level resets them.

Question: What exactly is victory, defeat, and checkpoint behavior?

Build answer: Use three compact encounters. Defeating every enemy in an encounter enables its next checkpoint; walking into that checkpoint activates it and permits progress and equipment changes. The final checkpoint reached after every enemy is defeated is victory. Zero HP is defeat followed by checkpoint respawn. Activating a checkpoint does not restore health; Healing Relics and respawning restore health as defined above. All checkpoint details beyond the handoff are declared implementation assumptions.

Part 2 - Story pitches (40 points)

Each member contributed one story pitch. Each pitch gives a reason for all four mechanics and identifies a clue the player finds in the world.

Pitch 1 - Rocky Clarke: The Last Shift

Premise: A worker returns to an evacuated underground facility to find a missing sibling and discovers that the security system is still enforcing the final evacuation lockdown.

Mechanic	Why the player does it
Utility / Inventory	The ten-slot harness limits armor and Healing Relics; emergency medical capsules left behind by workers help the player survive the search.
Armor	Abandoned workers' suits protect against the security units, with mobility and protection affecting which route is practical.
Shifting Environment	Emergency partitions still cycle every 30 seconds; the maintenance flashlight holds a photo-sensitive safety panel while the worker passes.
Enemies	Light and Heavy security units attack anyone moving through the sealed sectors; disabling every unit releases the next checkpoint.

Something found: The sibling's distinctive orange scarf first appears in a shared photograph, then is seen tied to a fence beyond the final evacuation exit. Footprints continue away from it, suggesting escape rather than death.

Pitch 2 - Kai Harris-Coles: The Museum Below

Premise: A conservator enters a sealed underground museum and reconstructs a failed evacuation from the exhibits and staff belongings left behind.

Mechanic	Why the player does it
Utility / Inventory	The conservator balances limited space for protective suits and Healing Relics recovered from abandoned museum first-aid supplies.
Armor	Old exhibition-handling suits provide different protection and mobility through security-controlled galleries.
Shifting Environment	Automated exhibit partitions continue rearranging galleries; a work light pauses one safety sensor.
Enemies	Light and Heavy museum security units remain hostile and must be disabled to open each security checkpoint.

Something found: An empty child-sized coat beside a staff badge and a line of disturbed display objects leads toward a service exit, showing that someone abandoned the collection to help a visitor.

Part 2 - Remaining pitches / Part 3 - Selection

Pitch 3 - Zion King: After the Last Train

Premise: A stranded commuter searches an evacuated transit station for evidence of where the last passengers went while automated defenses keep the station sealed.

Mechanic	Why the player does it
Utility / Inventory	A ten-slot emergency pack forces choices between protective suits and Healing Relics abandoned at evacuation stations.
Armor	Transit maintenance suits trade speed for protection against station security.
Shifting Environment	Safety barriers cycle between evacuation routes, and a maintenance flashlight pauses one barrier.
Enemies	Light and Heavy security units enforce the lockdown; clearing each sector unlocks its next checkpoint.

Something found: A trail of scattered luggage ends at a stroller parked beside an open evacuation passage, with many footprints continuing outside.

Pitch 4 - Michael Nearn: A Signal Left Behind

Premise: A technician follows a repeating distress tone through an abandoned relay station and discovers the route an earlier rescue crew used to escape.

Mechanic	Why the player does it
Utility / Inventory	A limited harness holds protective suits and Healing Relics from the rescue crew medical supplies.
Armor	Relay maintenance suits offer different protection against hostile security equipment.
Shifting Environment	Signal-shielding partitions cycle automatically; the flashlight holds a safety sensor during traversal.
Enemies	Light and Heavy relay security units block access to successive checkpoints and must be disabled.

Something found: The repeating tone comes from an abandoned rescue beacon facing an open service exit, with a crew emblem caught on the outside fence.

Part 3.1 - Story selected (10 points)

Selected story: The Last Shift. Its evacuation setting naturally explains every received mechanic, while the recurring scarf provides a personal story the player can infer from objects and spaces.

Why the other pitches lost:
The Museum Below: The artifacts could imply a collection objective that the received mechanics do not actually require.
After the Last Train: Making a train departure central could imply a time limit or boarding mechanic that the handoff does not include.
A Signal Left Behind: Following a signal could imply a tracking or radio interaction that the handoff does not include.

Chosen story - How the world tells it

Core loop

Choose equipment at a checkpoint, navigate cycling partitions with the flashlight, defeat the remaining security units, inspect the evidence left along the route, and reach the next checkpoint while managing the ten-slot inventory.

Beginning, middle, and ending

Beginning: The opening area contains a photograph of two siblings; one wears an orange scarf. Two adjacent work stations repeat the same personal objects from the photograph. One station has been hastily abandoned. The player has returned to search for that person, without a quest giver or dialogue scene.

Middle: Three connected sectors show an interrupted evacuation. A meal remains on a table, a work suit lies beside a partition, chairs are pushed away from the alarms, and scuff marks lead toward the service route. Personal objects recur near the route to the next checkpoint. Functional loot uses the received pickup rules; background props are observed without a new interaction button.

Ending: Beyond the third cleared sector, the final checkpoint opens onto an evacuation exit. The orange scarf is tied outside the fence and footprints continue into daylight. The player finds evidence that the sibling made it out. Victory still depends on the received enemy-clearance and checkpoint rules, not on collecting a special story item.

Why each mechanic matters to the selected story

Utility: The abandoned facility contains more equipment than the worker can carry. Armor competes with emergency Healing Relics for space, making the supplies left behind meaningful to survival.
Armor: Different uniforms reveal the jobs people held, while their protection and movement values shape survival.
Environment: The evacuation architecture still moves after its operators have gone; the flashlight exploits its old safety response.
Enemies: The security units continue obeying the lockdown, creating the obstacle between the worker and the truth about the evacuation.

Found theme and text budget

Tell the story with repeated personal objects, empty work spaces, directional debris, alarm sounds, footsteps, and the contrast between confinement and daylight. No collectible journal, dialogue tree, quest log, key hunt, or new story-solving action is required.

Planned in-world words: "EVACUATE" (1), "ASSEMBLY POINT" (2), "WAIT FOR ME OUTSIDE" (4). Total environmental story text: **7 words**. Use no spoken dialogue. Reserve up to 23 additional words for necessary HUD and result text, for a planned total of at most **30 counted words**, comfortably below 150. Title-screen and controls text are excluded by the assignment; count every other visible label, sign, item name, and spoken word before submission.

Scope and visual direction

Build a compact single-player 3D game with three short encounters, two enemy types, three armor types, ten inventory slots, and two authored positions per moving obstacle. Use simple industrial geometry, slate-gray surfaces, cyan safety lighting, red alarm lighting, and one orange scarf motif. No multiplayer, crafting, armor durability, abilities beyond the four assigned mechanics, or extra enemy classes. Healing Relics are implemented within Utility / Inventory using the defined R-input consumption rules.

Part 3.2 - Mechanic ownership (10 points)

Member / mechanic	Owned state and responsibilities	Completion check
Rocky Clarke Shifting Environment	Own the global 30-second cycle, obstacle positions, active freeze target, 10-second freeze, 5-second cooldown, flashlight targeting, collision checks, and safe path validation.	F works within 10 feet, affects only one object, cannot refresh early, and unsafe shifts never damage or trap the player.
Kai Harris-Coles Enemy System	Own Light/Heavy HP, detection, chase, leash, attacks, player weapon damage handling, guaranteed drops, and remaining-enemy counts.	HP, damage, speeds, attack timings, 6/2/10-unit ranges, wall avoidance, drops, and checkpoint gating match the documented rules.
Zion King Armor Types	Own equipped suit state, suit modifiers, physical/elemental damage calculation, speed modifier, knockback immunity, and backup-armor death behavior with Utility.	Only one suit is active; all numerical modifiers work; equipped armor survives death; backup suits drop correctly.
Michael Nearn Utility / Inventory	Own the 10-slot container, item costs, pickup acceptance, checkpoint-only equipment permissions, item removal/drop transactions, and relic storage, R-input consumption, 25-HP healing, and the 5-second use cooldown.	Equipped and backup items count; oversized pickups stay in the world; swaps and voluntary drops fail outside checkpoints.

Where the four systems connect

Utility to Armor: Inventory owns item possession and slot usage; Armor owns which suit is active. One validated equipment transaction updates them together.
Armor to Enemies: Enemies submit base damage and damage type; Armor computes the final damage and knockback permission. Apply armor exactly once.
Enemies to Utility: Each enemy defeat spawns exactly one item; Utility alone accepts or rejects the pickup and changes inventory contents.
Environment to Enemies: Obstacles expose updated blocking geometry; enemy navigation changes with it. Freezing a wall can preserve an escape route or keep a threat separated.

Integration risk and prevention

The first likely failure is an equipment pickup or death changing slot usage differently from the equipped suit. Michael and Zion will define shared item IDs and one transaction path before building UI. Rocky and Kai will test enemy pursuit through both obstacle layouts and through frozen mixed layouts before adding story assets.

Development schedule

October 23: Complete the story selection, build interpretations, repo setup, and individual identity-file commits.
October 26: Prepare the story checkpoint, reuse Part 1 as its section 5, and include the resolved handoff interpretations as section 5.
October 27-29: Each owner implements and verifies their mechanic in a small test scene.
October 30: Combine all four systems into one playable graybox.
November 1-2: Add environmental storytelling and run teammate playtests; attend the class final playtest on its announced date.
November 3: Fix interpretation and integration issues; audit words and export.
November 4: Verify the final build and submit under the full Project 4 rules.

Part 3.3 - Repo setup and submission (10 points)

Public team repository: https://github.com/notziking8/Project-4.git
Rocky's commit hash: 2272fc95b1b238dd42dcacaa95d10c8641b62142

Our team has set up the public repository with the Godot project, selected the Compatibility renderer, and added all members as collaborators. Each member completed the required named-file commit, checked that their Git email matches their GitHub account, and pulled before pushing. My repository link and commit hash are included at the top of this submission.

Requirement	Completed setup
Public repo containing Godot project	Completed; repository linked above.
Compatibility renderer	Selected in the Godot project.
Other members added as collaborators	All team members have been added.
Every member pushes a commit by 11:59 p.m.	Completed; my full commit hash is provided above.
First commit contains a named one-line file	Completed using the member-specific files listed below.
Check email; pull before push	Each member checked their Git email and pulled before pushing.

Required identity files

Member	New file	One-line content
Rocky Clarke	Rocky_Clarke.txt	Rocky Clarke - Shifting Environment
Kai Harris-Coles	Kai_Harris-Coles.txt	Kai Harris-Coles - Enemy System
Zion King	Zion_King.txt	Zion King - Armor Types
Michael Nearn	Michael_Nearn.txt	Michael Nearn - Utility / Inventory

Work organization

We use the same Godot version, exclude .godot/ from version control, and keep each mechanic in its owner's scenes and scripts. We coordinate before editing another owner's files.

Mechanics source: Team 2 Mechanics for Team 3.pdf. Part 1 records the team decisions used to resolve ambiguous or unspecified rules.